
EarStudio:

State-of-the-art Bluetooth Receiver

Technologies and Main Features

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EarStudio is a compact Hi-Fi device designed to deliver superior sound quality in every aspect, including

- **Special back-end DSP algorithm and firmware**
- **Special front-end output HW architecture**

The primary issue for the current Bluetooth codecs (SBC, AAC, aptX) is the 16bit limited resolution, supporting CD-like quality. The new aptX-HD is the 24bit codec, but there's not yet many aptX-HD mobile phones released in the market. At this moment, we are experiencing almost every Bluetooth audio streaming with 16bit resolution. Moreover, the jitter caused by inter-device reference clock mismatch also makes the Bluetooth sound quality worse than the wired interface.

Driving headphones and earphones ranged from 16Ω to 600Ω efficiently is another important issue for Bluetooth receivers. However, most Bluetooth receivers in the market have the limitation of the analog front-end, eventually making another loss and distortion to the sound over Bluetooth codec.

Equipped with **CSR 8675** and 2 DACs of **AK4375a**, EarStudio places the most focus on the sound quality along with convenience and affordability. EarStudio reproduces sound over Bluetooth into 88.2KHz/24bit, leveraging Radsone's patented DCT technology, which reduces subtle digital noise and cleans the jitter from Bluetooth streaming. Also, Radsone's patented DualDrive architecture enables the precise reproduction of analog outputs into any headphones or earphones. Meaning, EarStudio has the capability of delivering high-quality sound that is much closer to analog sound even through Bluetooth.

CSR8675 is the most advanced Bluetooth platform in the current market. It has MCU and DSP(Digital Signal Processor) which allows support for 24bit audio over Bluetooth with aptX HD. As we are one of an official member of CSR eXtension Partner program, we've been developing many Bluetooth headsets and receivers using CSR8675, since after the birth of CSR8675.

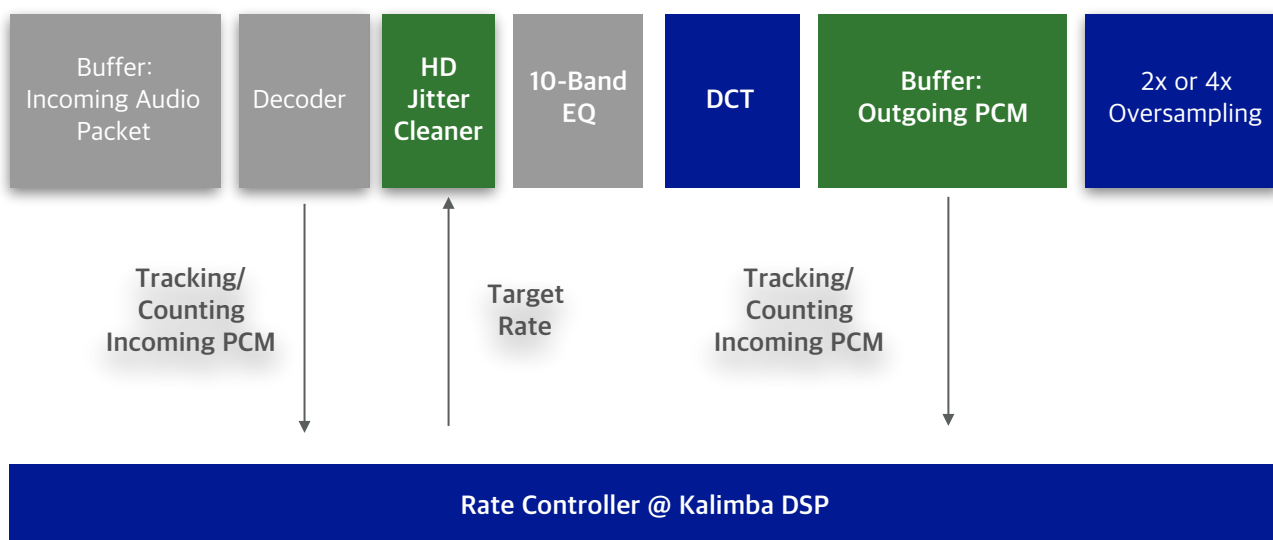


Fig 1. Audio Framework with Radstone DCT Blue sound engine

Although CSR8675 firmware stack delivers the better sound quality than the other solution, we've added our own DSP solution, named **DCT Blue** (DCT : Distinctive Clear Technology), into CSR8675, for the less jitter and the less noise with Bluetooth A2DP.

Fig.1 shows the audio framework with DCT Blue sound engine. The blue blocks are newly designed and implemented, and the green blocks are modified and optimized, all by Radstone. The Radstone's optimized Bluetooth audio framework, running on CSR8675 Kalimba DSP, provides the better sound than any other Bluetooth devices can do.

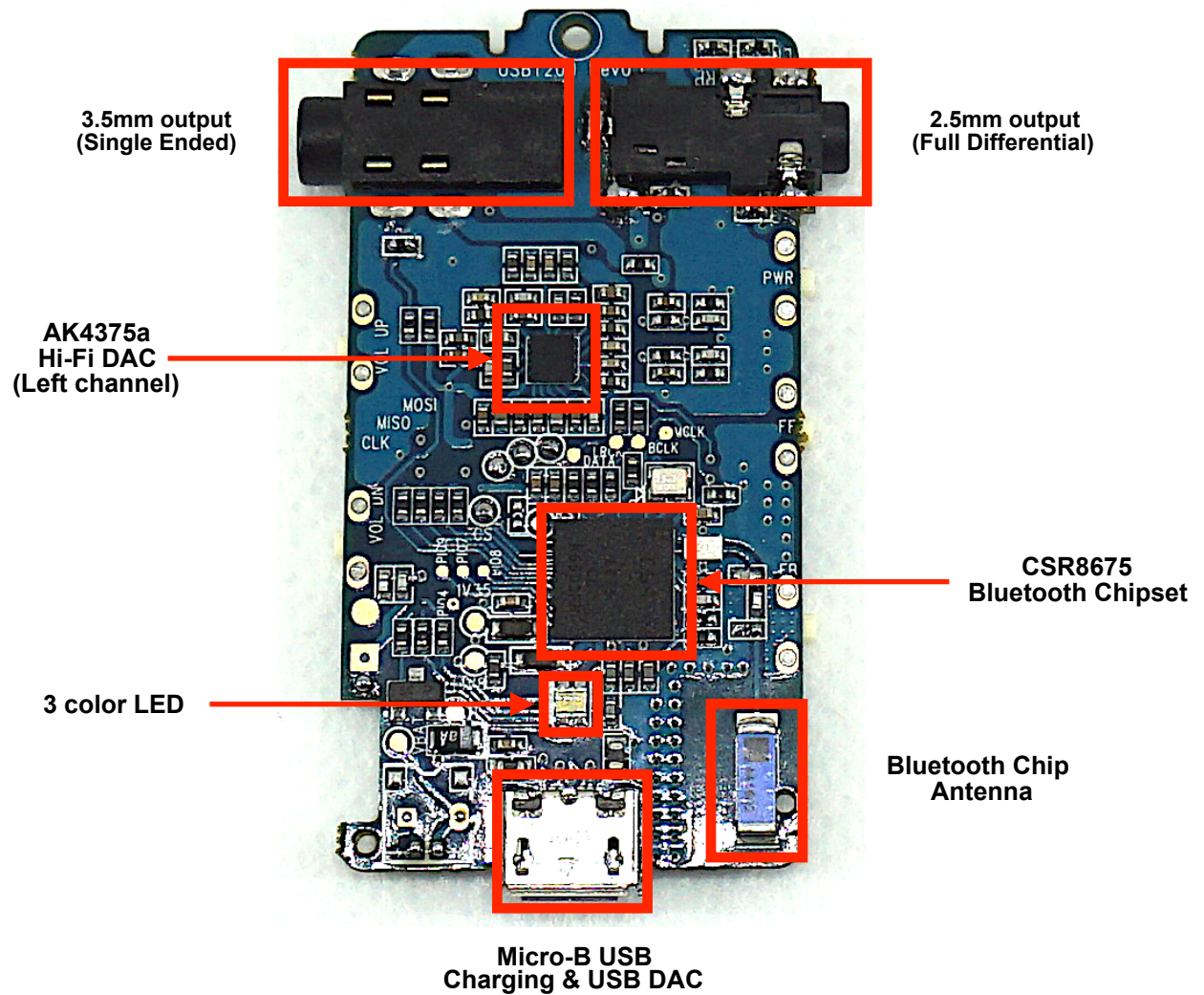


Fig 2. EarStudio PCB Assembly - Top

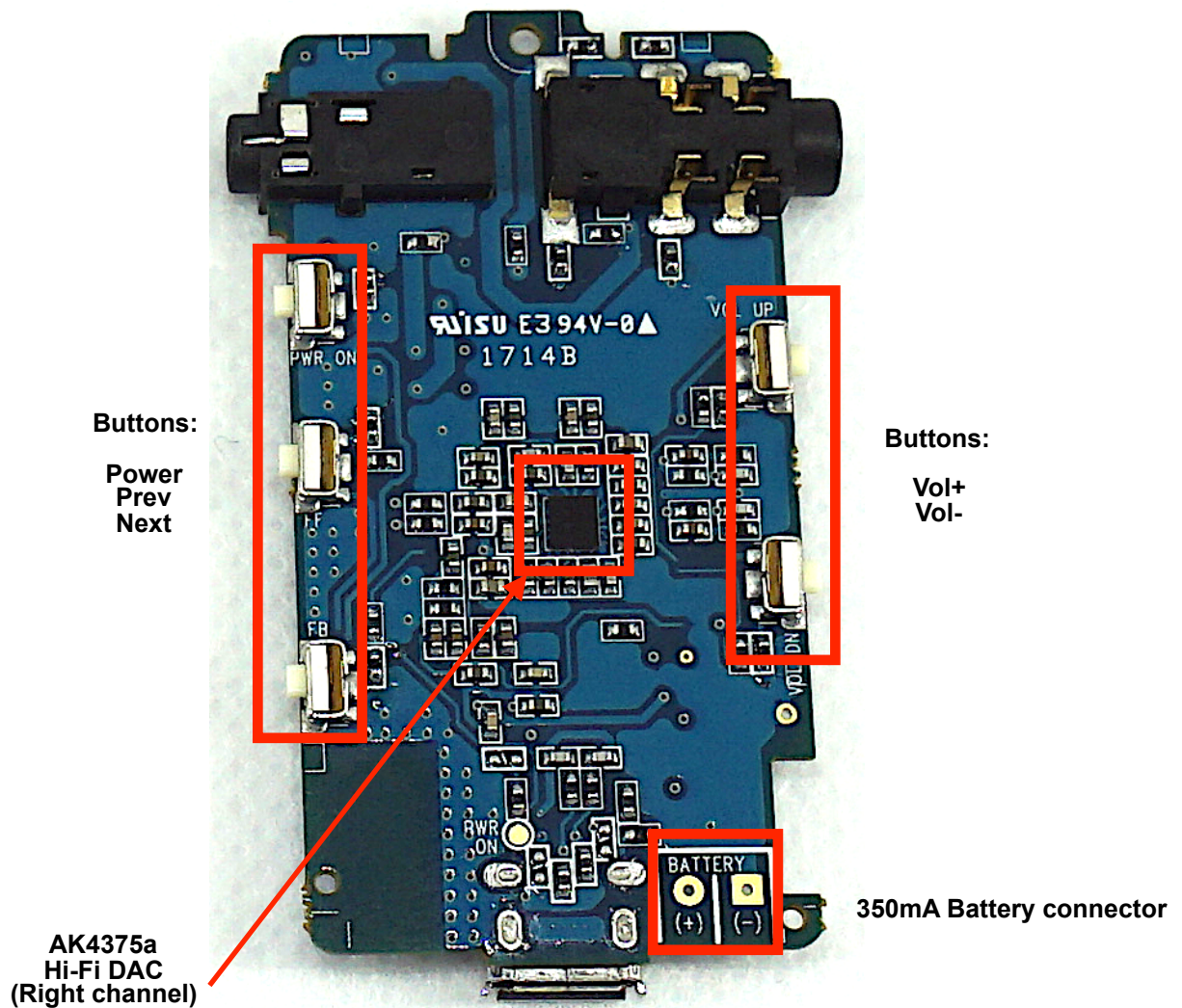


Fig 3. EarStudio PCB Assembly - Bottom

EarStudio has TWO High-Quality DACs (AK4375a), unlike other Bluetooth Receivers, to deliver the best sound performance. AK4375a, which was integrated into EarStudio, has built-in headphone amplifier with the analog gain controller (i.e. Programmable Gain Amplifier).



In general, dual DAC architecture provides the better inter-channel isolation (a.k.a. cross-talk), enabling the better channel separation and stereo imaging. Moreover, the proprietary DualDrive technology offers a unique competitive advantage of doubling HW performance in comparison to other dual DAC architectures.

For 3.5mm single-ended output, the **DualDrive** technology reduces the output impedance by half and gives twice much current than one single AK4375a can deliver. For 2.5mm full-differential output, the voltage is increased by twice.

As illustrated in Fig.2 and Fig.3, Radsone proprietary DualDrive architecture has NO other additional components in the way from the amplifier to the output, such as switching IC, which degrades the sound quality. It enables to deliver the pure signal without any loss or distortion, and with enough power. For more detailed information, please refer to the DualDrive technology in References.

Moreover, EarStudio is the first Bluetooth audio product which adopted **Analog Volume Control**. It maintains the better sound quality even at medium or low volume levels.

Analog volume control works through volume knobs of dedicated control apps or hardware buttons on the device. For more detailed information, please refer to the Analog Volume Control in References.

To utilize the USB functionality of CSR8675, EarStudio provides USB Audio Class 1.0 as a generic USB DAC. Unfortunately, the hardware capability of CSR8675 USB limits the USB streaming quality as 48KHz/16bit, but the same analog front-end and the same DCT Blue engine provides much better sound than any other conventional 48KHz/16bit USB DAC can deliver.

EarStudio secures 14 hours of continuous playback by 350mAh battery and optimized system design. Typical Bluetooth receivers usually have much shorter playback time, mostly under 10 hours. Even with its extensive battery life, EarStudio only weighs 20g, lighter than one AA battery (24g).

Also, DFU(Device Firmware Upgrade) over USB is another benefit of EarStudio providing easy firmware upgrade and bug fixes with a PC or laptop.

Lastly, EarStudio provides a dedicated remote control application both for iOS and Android devices. Any users can enjoy the full degree of freedom to control the full functionalities of EarStudio's hardware and firmware according to their preferences.

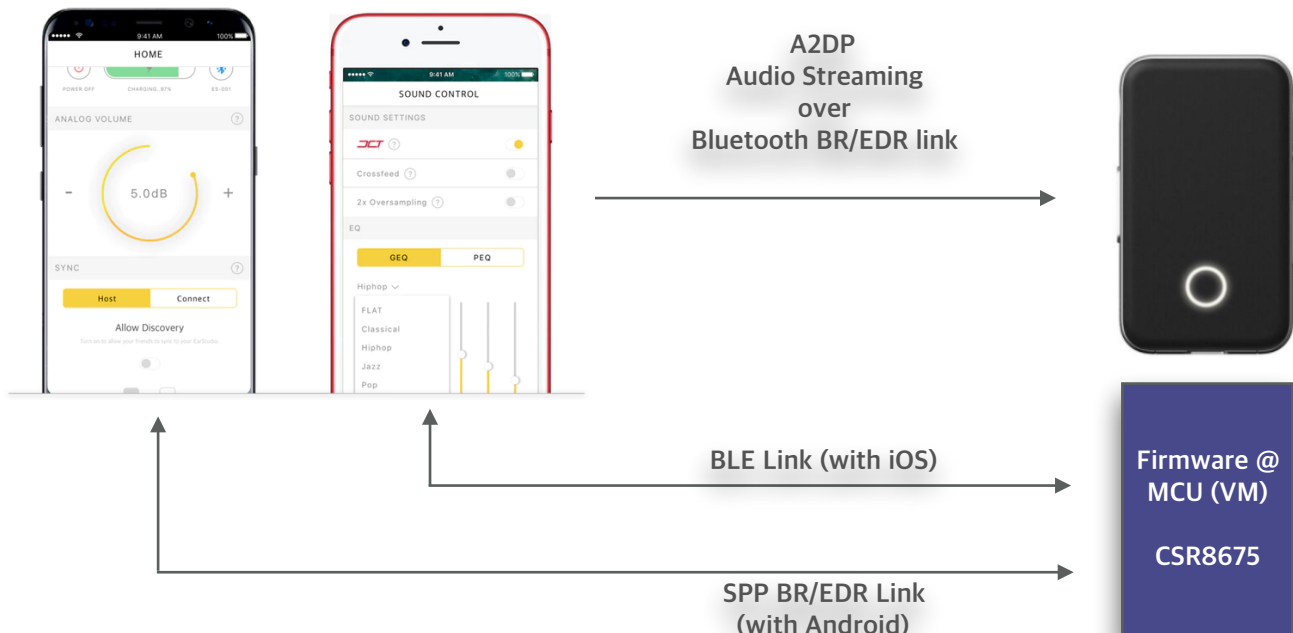


Fig.4. EarStudio mobile application

To summarize, EarStudio was developed based on years of extensive research and engineering efforts for the highest quality of Bluetooth audio products. EarStudio is a true compact Hi-Fi device in the Bluetooth audio market, powered by Radsone's proprietary technologies to offer superior sound quality and convenience all at once.

References

1. **DualDrive Technology**
2. **Analog Volume Control**
3. **Bluetooth / USB Performances**
4. **Radstone IP**